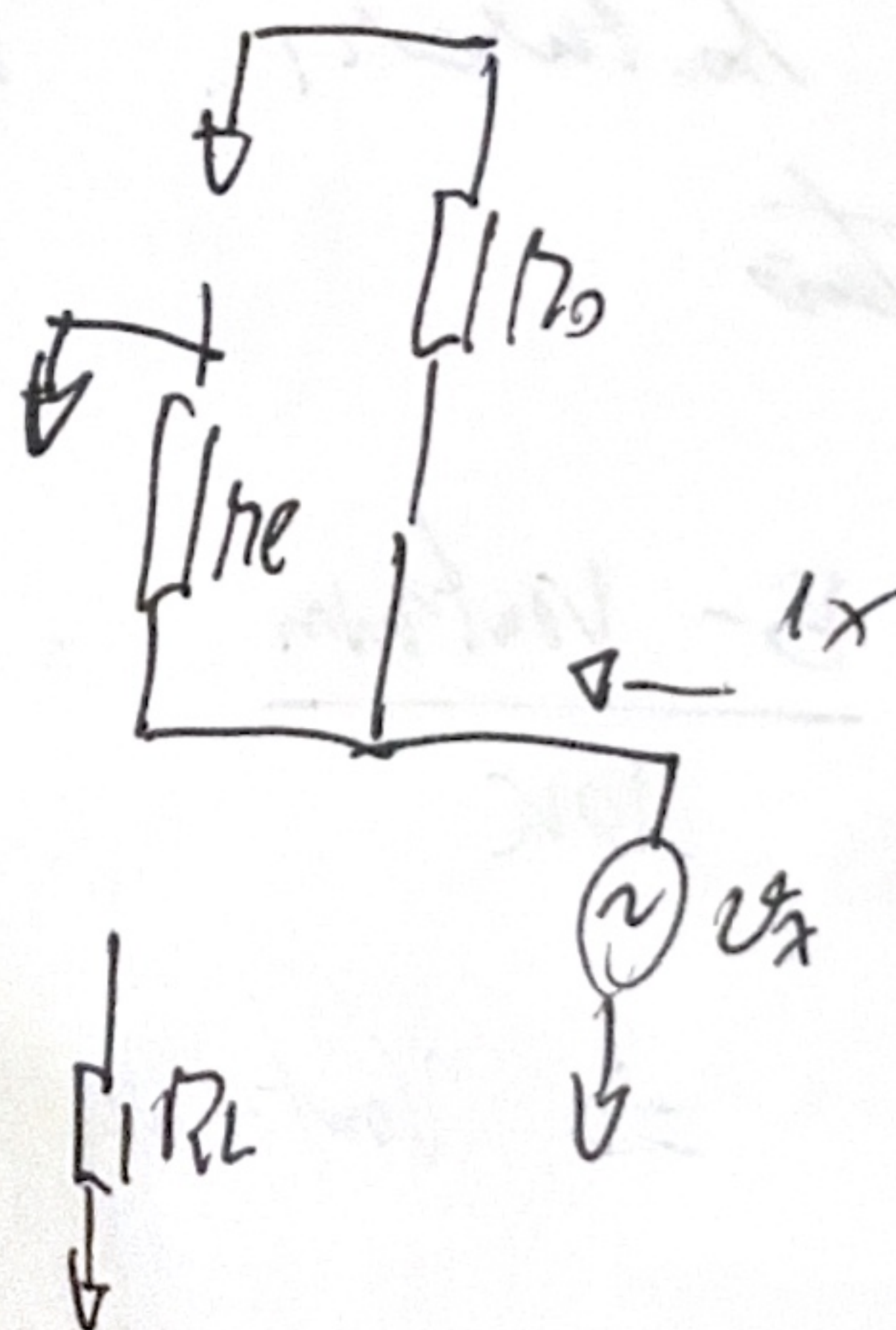
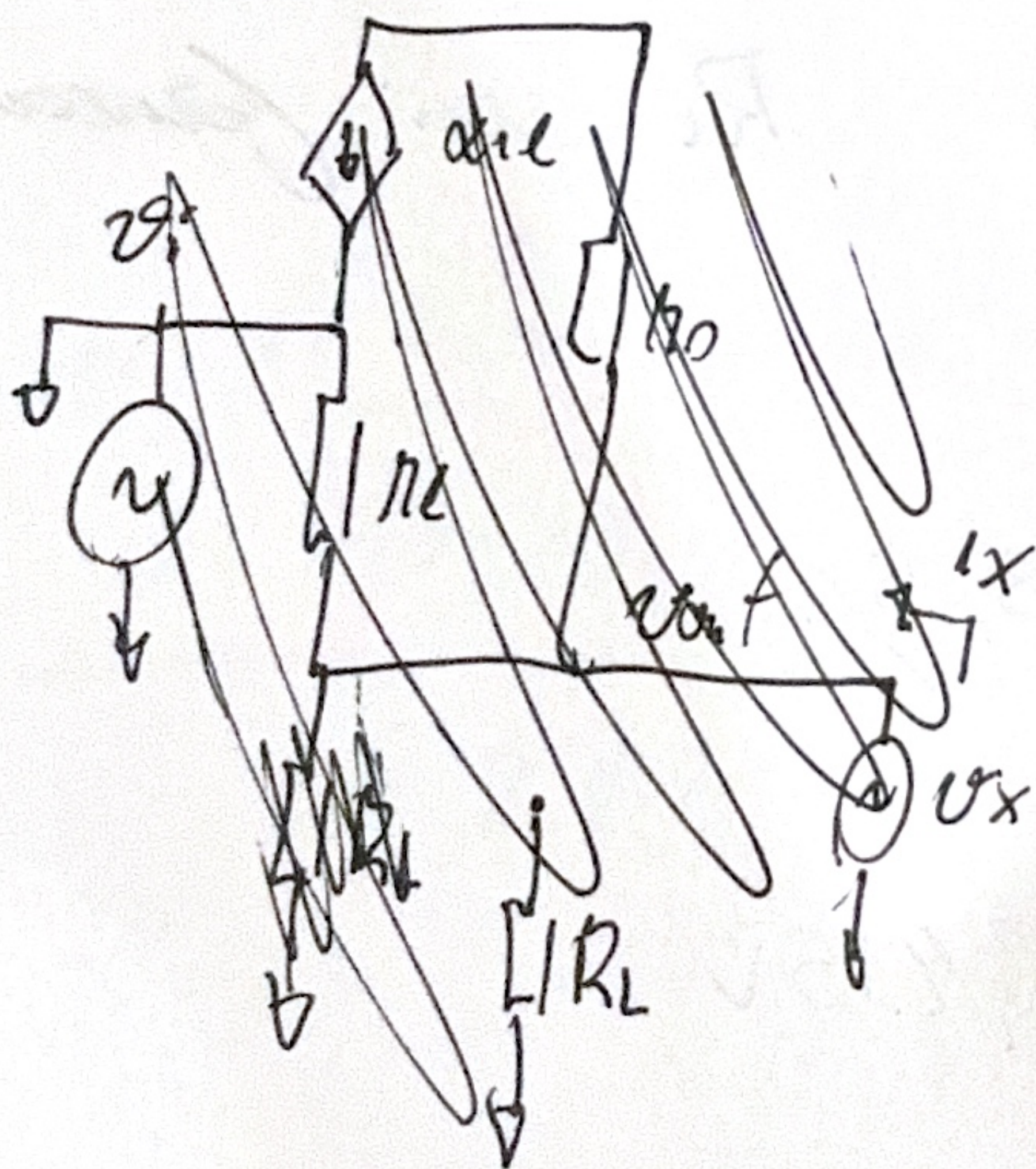
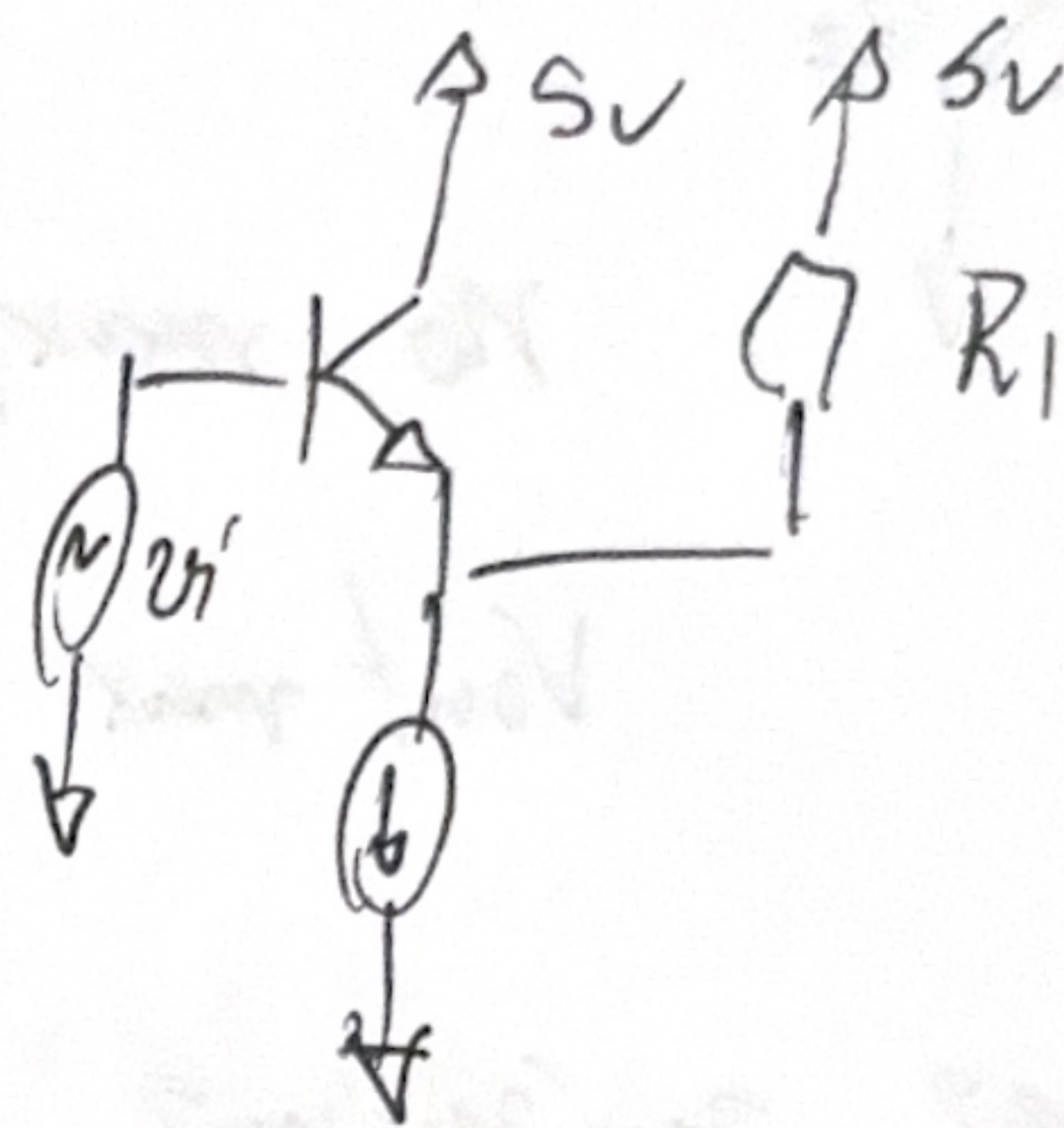
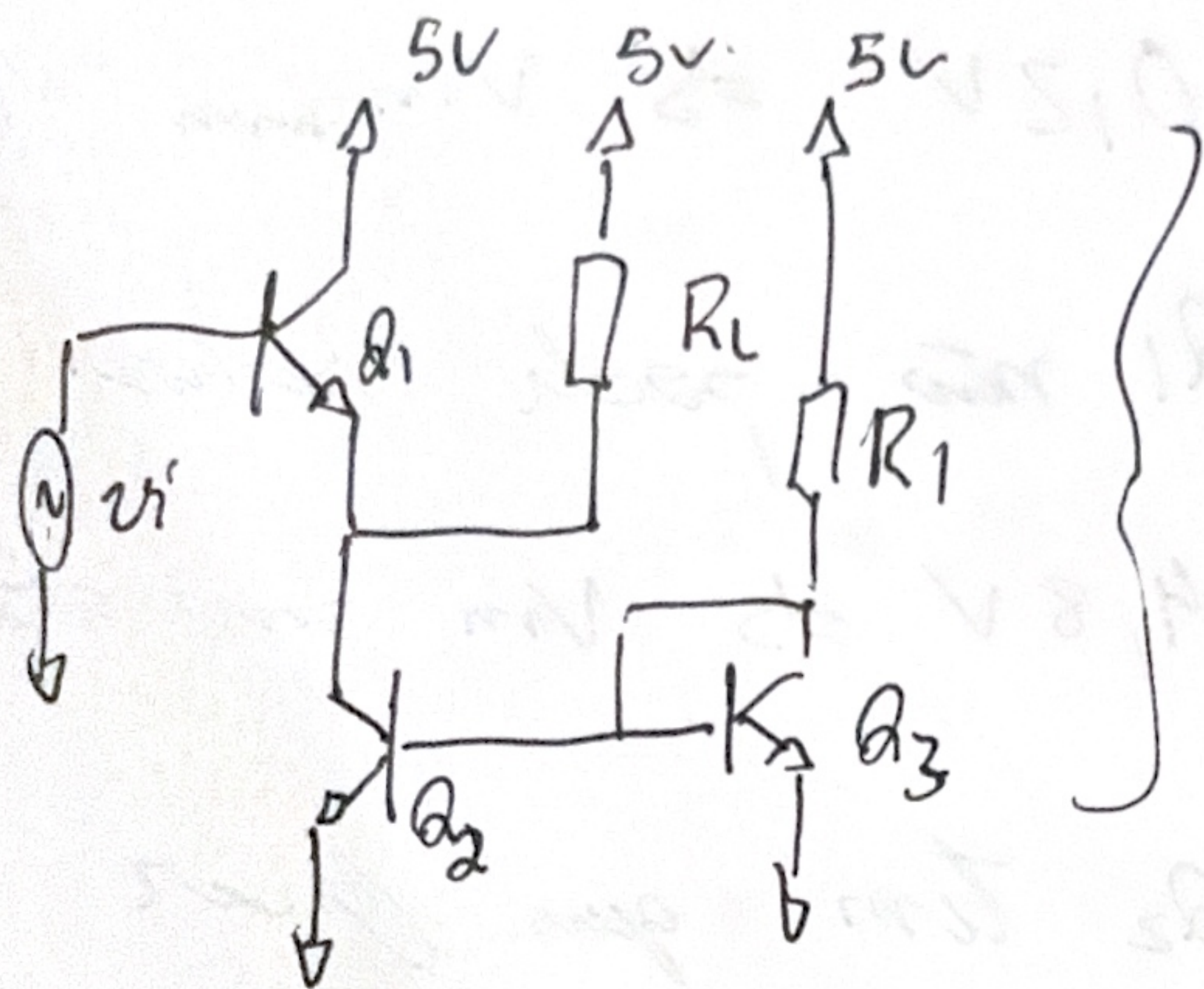


⑧ β e V_T altos - no muito elevado e I_B baixa.

* Super transistores iguais

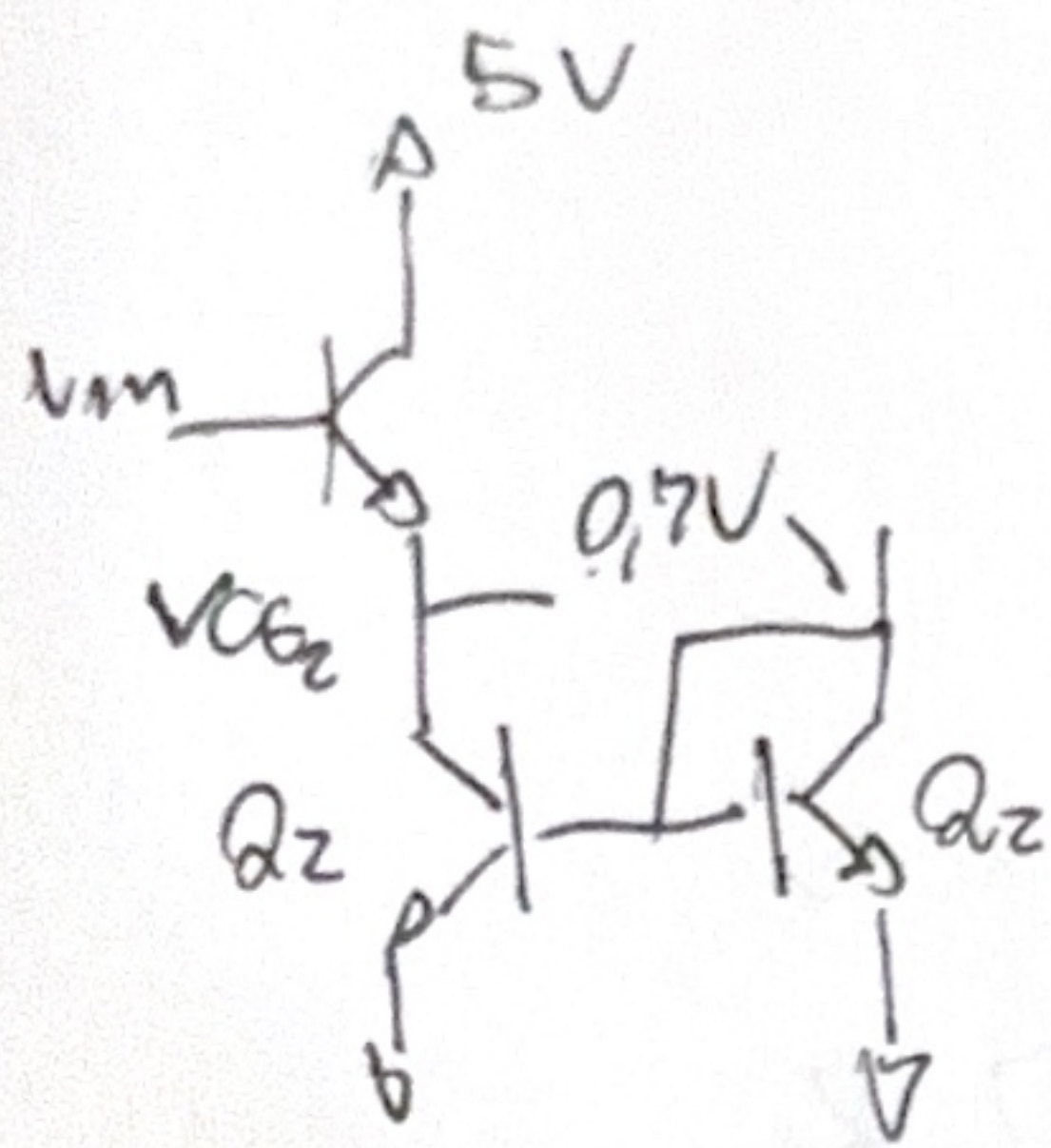


$$Z_x = \frac{v_x}{i_x} = r_o \parallel r_e \approx r_e$$

$$r_e = \frac{\alpha}{g_m} = \frac{\beta}{\beta+1} \cdot \frac{nV_T}{I_C} \approx \frac{nV_T}{I_C} = \frac{0,02585}{I_C} = 100$$

$$I_C = 258,5 \mu A \Rightarrow R_1 = \frac{5 - 0,7}{258,5 \mu} = 16,634 K$$

2 $R_L \rightarrow \infty$



No mínimo, R_2 não pode saturar, então

$$V_{out\min} = 0,2V \Rightarrow V_{in\min} = 0,9V$$

No max, Q_1 não pode saturar:

$$V_{out\max} = 4,8V \Rightarrow V_{in\max} = 5,5V$$

3 Agora, no mínimo, Q_2 tem que ficar ligeiramente saturar, mas R_1 vai fornecer corrente, então

~~$\frac{5 - V_{out\min}}{258,5\mu}$~~ $\frac{5 - V_{out\min}}{10K} = 258,5\mu$

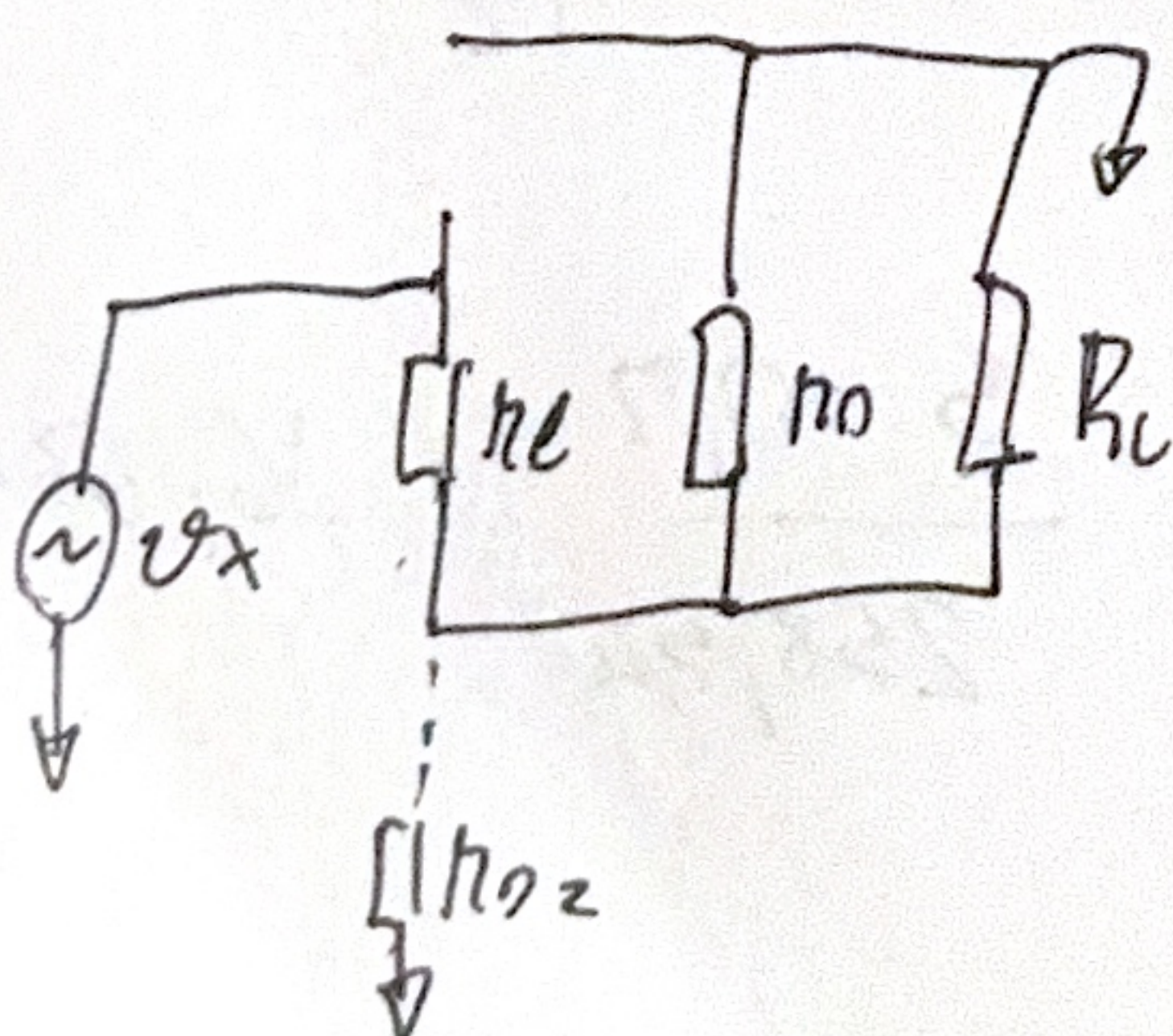
$$\Rightarrow V_{out\min} = 2,415V$$

$$V_{in\min} = V_{out\min} + 0,7 = 3,115V$$

O limite superior fica igual

Olhando pela base com $\beta+1$

4



$$\begin{aligned} Z_{in} &= (\beta+1)(r_{\pi} + R_E || R_L) \\ &= 101 \left(\frac{100}{101} \frac{25,85m}{258,5\mu} + 1K || \frac{100}{258,5\mu} \right) \\ &= 101 \cdot 110,84K \end{aligned}$$